K-means formulation of the block-group clustering

(`standardize.py` + `cluster.py`, default settings)

1 The textbook objective

Given observations $\mathbf{x}_1, \ldots, \mathbf{x}_n \in \mathbb{R}^d$, k-means seeks

$$\arg\min_{\mathbf{S}} \sum_{i=1}^{k} \sum_{\mathbf{x} \in S_i} \left\| \mathbf{x} - \boldsymbol{\mu}_i \right\|^2 = \arg\min_{\mathbf{S}} \sum_{i=1}^{k} |S_i| \operatorname{Var} S_i, \qquad \boldsymbol{\mu}_i = \frac{1}{|S_i|} \sum_{\mathbf{x} \in S_i} \mathbf{x}. \tag{1}$$

The rest of this document defines exactly what n, d, $\mathbf{x}$, $\boldsymbol{\mu}_i$ and $\| \cdot \|$ are in our pipeline.

2 Index sets

- $\mathcal{B}$: all 2020 Census block groups (BGs) in the ACS 2019–23 universe, 50 states + DC; $|\mathcal{B}| = 239{,}741$. Each $b \in \mathcal{B}$ has a 12-character GEOID; $\mathrm{st}(b)$ is its first 2 characters (state FIPS).
- Row filter (the *fit set*):

$$\mathcal{F} = \{\, b \in \mathcal{B} : \ \mathrm{pop}_b \geq 50 \text{ and } \mathrm{households}_b > 0 \,\}, \qquad |\mathcal{F}| = 236{,}829.$$

- Region R (a set of state FIPS codes chosen with `--states`):

$$\mathcal{B}_R = \{\, b \in \mathcal{F} : \ \mathrm{st}(b) \in R \,\}, \qquad n = |\mathcal{B}_R|.$$

  For $R = \{09, 34, 36\}$ (CT, NJ, NY): $n = 24{,}680$.
- Features $\mathcal{J}$, $d = |\mathcal{J}| = 42$ (Table 1).
- Concepts $\mathcal{G}$, $|\mathcal{G}| = 15$: each feature belongs to exactly one concept $g(j) \in \mathcal{G}$; $n_g = |\{j : g(j) = g\}|$ is the concept's column count.
- $\mathcal{E} = \{\texttt{ecef_x_km}, \texttt{ecef_y_km}, \texttt{ecef_z_km}\} \subset \mathcal{J}$ is the *distance* concept ($n_{\text{distance}} = 3$).
- Pooled-scale blocks $\mathcal{P} = \{\mathcal{E}, K_{\text{race}}, K_{\text{units}}, K_{\text{lang}}, K_{\text{occ}}\}$: disjoint sets of columns that share one scale (Steps 3–4). K_{race} is the 7 race shares, K_{units} the 6 home-type shares, K_{lang} the 5 tract language shares (not `lep_hh_share`), and K_{occ} the 5 occupation shares. Every other feature is scaled on its own.

3 Constructing $\mathbf{x}_b$

3.1 Step 0: raw values

$r_{b,j}$ is the raw value of feature j for BG b (possibly missing). The three distance features are Earth-Centered Earth-Fixed coordinates of the BG's population-weighted center (latitude φ_b, longitude λ_b; TIGER internal point as fallback) on the WGS84 ellipsoid at height 0:

$$a = 6378.137 \text{ km}, \qquad f = \tfrac{1}{298.257223563}, \qquad e^2 = f(2-f), \qquad N(\varphi) = \frac{a}{\sqrt{1 - e^2 \sin^2 \varphi}},$$

$$\mathbf{p}_b = \begin{pmatrix} r_{b,\mathrm{x}} \\ r_{b,\mathrm{y}} \\ r_{b,\mathrm{z}} \end{pmatrix} = \begin{pmatrix} N(\varphi_b) \cos\varphi_b \cos\lambda_b \\ N(\varphi_b) \cos\varphi_b \sin\lambda_b \\ N(\varphi_b)\,(1-e^2) \sin\varphi_b \end{pmatrix} \text{ km}.$$

3.2 Step 1: per-feature transform

With $\mathrm{clip}(v; \ell, u) = \min(\max(v, \ell), u)$,

$$t_{b,j} = T_j(r_{b,j}), \qquad T_j(v) = \begin{cases} v & \text{none} \\ \ln\big(1 + \max(v, 0)\big) & \text{log1p} \\ \mathrm{clip}(v; \ell_j, u_j) & \text{winsor} \\ \ln\big(1 + \max(\mathrm{clip}(v; \ell_j, u_j), 0)\big) & \text{winsor_log1p} \end{cases}$$

The winsor bounds are empirical quantiles of the observed raw values over the national fit set,

$$\ell_j = \widehat{Q}_{\mathcal{F}}\big(r_{\cdot,j};\, p_j^{\mathrm{lo}}\big), \qquad u_j = \widehat{Q}_{\mathcal{F}}\big(r_{\cdot,j};\, p_j^{\mathrm{hi}}\big), \qquad (p_j^{\mathrm{lo}}, p_j^{\mathrm{hi}}) = \begin{cases} (0.020,\, 0.995) & j = \texttt{violent_crime_rate} \\ (0.005,\, 0.995) & \text{otherwise.} \end{cases}$$

Fitted values are listed in Table 1.

3.3 Step 2: hierarchical median imputation

Let $O_j \subseteq \mathcal{F}$ be the BGs where $t_{b,j}$ is observed. For a BG b define nested geographic neighbourhoods by GEOID prefix,

$$A_L(b) = \{\, b' \in \mathcal{F} :\ \mathrm{GEOID}_{b'}[{:}\kappa_L] = \mathrm{GEOID}_b[{:}\kappa_L] \,\}, \qquad (\kappa_{\mathrm{tract}}, \kappa_{\mathrm{county}}, \kappa_{\mathrm{state}}, \kappa_{\mathrm{nation}}) = (11, 5, 2, 0).$$

For a scalar feature,

$$\tilde{t}_{b,j} = \begin{cases} t_{b,j} & b \in O_j \\ \mathrm{med}\{\, t_{b',j} : b' \in A_{L^*}(b) \cap O_j \,\} & b \notin O_j, \end{cases}$$

where L^* is the first level in (tract, county, state, nation) whose donor set $A_L(b) \cap O_j$ is non-empty.

Compositional blocks $K \in \{$race (7), home type (6), tract language (5), occupation (5)$\}$ are imputed all-or-nothing: if any share in K is missing for b, the whole vector is replaced. Donors are BGs whose block is complete, $O_K = \bigcap_{j \in K} O_j$, and

$$\mathbf{m}_K(b) = \big(\mathrm{med}\{t_{b',j} : b' \in A_{L^*}(b) \cap O_K\}\big)_{j \in K}, \qquad \tilde{t}_{b,j} = \frac{m_{K,j}(b)}{\sum_{j' \in K} m_{K,j'}(b)} \quad (j \in K),$$

so imputed shares still sum to 1. (`lep_hh_share` is in the language concept but not in the tract-language block, so it is imputed as a scalar.)

3.4 Step 3: national standardization (`features_standardized.parquet`)

Population moments over the fit set (ddof = 0):

$$m_j = \frac{1}{|\mathcal{F}|}\sum_{b\in\mathcal{F}} \tilde{t}_{b,j}, \qquad s_j = \Big(\frac{1}{|\mathcal{F}|}\sum_{b\in\mathcal{F}} (\tilde{t}_{b,j} - m_j)^2\Big)^{1/2}.$$

Each column of a pooled-scale block $P \in \mathcal{P}$ is centered on its own mean, but all columns of the block are divided by one shared scale: the root-mean of the block's column variances.

$$s_P = \Big(\frac{1}{|P|}\sum_{j\in P} s_j^2\Big)^{1/2}, \qquad s_{[j]} = \begin{cases} s_P & j \in P \in \mathcal{P} \\ s_j & \text{otherwise,} \end{cases} \qquad z_{b,j} = \frac{\tilde{t}_{b,j} - m_j}{s_{[j]}}.$$

Fitted values: $s_{\mathcal{E}} = 862.33$ km, $s_{K_{\text{race}}} = 0.1736$, $s_{K_{\text{units}}} = 0.1816$, $s_{K_{\text{lang}}} = 0.1305$, $s_{K_{\text{occ}}} = 0.1283$. Each block keeps total variance $\sum_{j\in P} \mathrm{Var}(z_{\cdot,j}) = |P|$, the same as $|P|$ separately z-scored features. Within the block, that variance is split among the columns in proportion to each category's actual spread.

3.5 Step 4: regional re-standardization (`--scale regional`, default)

Restrict to $\mathcal{B}_R$ and standardize again, using regional moments:

$$\bar{z}_j^R = \frac{1}{n}\sum_{b\in\mathcal{B}_R} z_{b,j}, \qquad \sigma_j^R = \Big(\frac{1}{n}\sum_{b\in\mathcal{B}_R} (z_{b,j} - \bar{z}_j^R)^2\Big)^{1/2}, \qquad \sigma_P^R = \Big(\frac{1}{|P|}\sum_{j\in P} (\sigma_j^R)^2\Big)^{1/2} \quad (P \in \mathcal{P}),$$

$$\check{z}_{b,j} = \begin{cases} (z_{b,j} - \bar{z}_j^R)/\sigma_P^R & j \in P \in \mathcal{P} \\ (z_{b,j} - \bar{z}_j^R)/\sigma_j^R & \text{otherwise, } \sigma_j^R > 0 \\ z_{b,j} - \bar{z}_j^R \ (=0) & \text{otherwise, } \sigma_j^R = 0. \end{cases}$$

Proposition (the national step cancels)**.** *Because Step 3 is an affine map with positive slope, Step 4 equals standardizing the imputed, transformed values directly within the region, using the same pooled blocks. Write $\bar{t}_j^R$ and τ_j^R for the regional mean and population s.d. of $\tilde{t}_{\cdot,j}$, and define the* effective regional scale

$$\tau_{[j]}^R = \begin{cases} \tau_P^R = \Big(\frac{1}{|P|}\sum_{j'\in P} (\tau_{j'}^R)^2\Big)^{1/2} & j \in P \in \mathcal{P} \\ \tau_j^R & \textit{otherwise.} \end{cases}$$

Then

$$\check{z}_{b,j} = \frac{\tilde{t}_{b,j} - \bar{t}_j^R}{\tau_{[j]}^R} \quad \forall j, \qquad \textit{in particular} \qquad \begin{pmatrix} \check{z}_{b,\mathrm{x}} \\ \check{z}_{b,\mathrm{y}} \\ \check{z}_{b,\mathrm{z}} \end{pmatrix} = \frac{\mathbf{p}_b - \bar{\mathbf{p}}^R}{\tau_{\mathcal{E}}^R}, \quad \tau_{\mathcal{E}}^R = \Big(\tfrac{1}{3}\sum_{a\in\{\mathrm{x},\mathrm{y},\mathrm{z}\}} \mathrm{Var}_{\mathcal{B}_R}(p_{\cdot,a})\Big)^{1/2}.$$

For $R = \{09, 34, 36\}$: $\tau_{\mathcal{E}}^R = 94.88$ km (per-axis s.d. 126.6, 63.3, 83.5 km), $\tau_{K_{race}}^R = 0.1764$, $\tau_{K_{units}}^R = 0.2219$, $\tau_{K_{lang}}^R = 0.1416$, $\tau_{K_{occ}}^R = 0.1280$.

Proof. For every j, $z_{b,j} = (\tilde{t}_{b,j} - m_j)/s_{[j]}$, so $\bar{z}_j^R = (\bar{t}_j^R - m_j)/s_{[j]}$ and $\sigma_j^R = \tau_j^R/s_{[j]}$. Within a block P the divisor $s_{[j]} = s_P$ is common, so $\sigma_P^R = \tau_P^R/s_P$. Substituting gives the result. □

With `--scale national` instead, $\check{z}_{b,j} = z_{b,j}$ (no Step 4). Everything below still holds with $(\bar{t}_j^R, \tau_{[j]}^R)$ replaced by $(m_j, s_{[j]})$, except the equal-contribution result of Section 5.4.

3.6 Step 5: concept weights

$$w_j = \frac{c_{g(j)}}{\sqrt{n_{g(j)}}}, \qquad c_g = 1 \;\; \forall g \in \mathcal{G} \quad (\texttt{CONCEPT_WEIGHTS}), \qquad W = \mathrm{diag}(w_1, \ldots, w_{42}).$$

So $w_j = 1$ for single-column concepts, $1/\sqrt{2} \approx 0.7071$ for 2-column concepts, $1/\sqrt{3} \approx 0.5774$, $1/\sqrt{5} \approx 0.4472$, $1/\sqrt{6} \approx 0.4082$, $1/\sqrt{7} \approx 0.3780$.

3.7 Step 6: the observation vector

$$\mathbf{x}_b = W\check{\mathbf{z}}_b \in \mathbb{R}^{42}, \qquad x_{b,j} = w_j\,\check{z}_{b,j} = \begin{cases} \dfrac{c_{\text{distance}}}{\sqrt{3}} \cdot \dfrac{p_{b,a(j)} - \bar{p}^R_{a(j)}}{\tau^R_{\mathcal{E}}} & j \in \mathcal{E} \\[2ex] \dfrac{c_{g(j)}}{\sqrt{n_{g(j)}}} \cdot \dfrac{\tilde{t}_{b,j} - \bar{t}^R_j}{\tau^R_{[j]}} & j \notin \mathcal{E} \end{cases}$$

where $a(j) \in \{\mathrm{x}, \mathrm{y}, \mathrm{z}\}$ is the ECEF axis of feature j, and $\tau^R_{[j]}$ is the block-pooled scale for share columns and the feature's own s.d. otherwise.

4 The objective with our definitions substituted

The observations are $\{\mathbf{x}_b\}_{b \in \mathcal{B}_R}$, $n = |\mathcal{B}_R|$, $d = 42$, k from `--k` (default 5). The partition is $\mathbf{S} = \{S_1, \ldots, S_k\}$ of $\mathcal{B}_R$.

Centroids. Since $\mathbf{x}_b$ is linear in $\check{\mathbf{z}}_b$ and in $\tilde{\mathbf{t}}_b$,

$$\boldsymbol{\mu}_i = \frac{1}{|S_i|}\sum_{b \in S_i} \mathbf{x}_b = W\bar{\check{\mathbf{z}}}_i, \qquad \mu_{i,j} = \begin{cases} \dfrac{c_{\text{distance}}}{\sqrt{3}} \cdot \dfrac{\bar{p}_{i,a(j)} - \bar{p}^R_{a(j)}}{\tau^R_{\mathcal{E}}} & j \in \mathcal{E} \\[2ex] \dfrac{c_{g(j)}}{\sqrt{n_{g(j)}}} \cdot \dfrac{\bar{t}_{i,j} - \bar{t}^R_j}{\tau^R_{[j]}} & j \notin \mathcal{E} \end{cases}$$

with $\bar{\mathbf{p}}_i = \frac{1}{|S_i|}\sum_{b \in S_i} \mathbf{p}_b$ (the cluster's mean ECEF point) and $\bar{t}_{i,j} = \frac{1}{|S_i|}\sum_{b \in S_i} \tilde{t}_{b,j}$. (`centroids_z.csv` reports $W^{-1}\boldsymbol{\mu}_i = \bar{\check{\mathbf{z}}}_i$.)

Objective. Substituting into (1), the regional means cancel inside $x_{b,j} - \mu_{i,j}$, leaving

$$\begin{aligned} &\arg\min_{\mathbf{S}} \sum_{i=1}^{k} \sum_{b \in S_i} \left\|\mathbf{x}_b - \boldsymbol{\mu}_i\right\|^2 \\ &= \arg\min_{\mathbf{S}} \sum_{i=1}^{k} \sum_{b \in S_i} \sum_{g \in \mathcal{G}} \frac{c_g^2}{n_g} \sum_{j:\, g(j)=g} \left(\check{z}_{b,j} - \bar{\check{z}}_{i,j}\right)^2 \\ &= \arg\min_{\mathbf{S}} \sum_{i=1}^{k} \sum_{b \in S_i} \left[\frac{c^2_{\text{distance}}}{3\,(\tau^R_{\mathcal{E}})^2} \left\|\mathbf{p}_b - \bar{\mathbf{p}}_i\right\|_2^2 + \sum_{g \neq \text{distance}} \frac{c_g^2}{n_g} \sum_{j:\, g(j)=g} \left(\frac{\tilde{t}_{b,j} - \bar{t}_{i,j}}{\tau^R_{[j]}}\right)^2 \right] \qquad (2) \\ &= \arg\min_{\mathbf{S}} \sum_{i=1}^{k} |S_i| \,\mathrm{Var}\, S_i, \qquad \mathrm{Var}\, S_i = \frac{1}{|S_i|}\sum_{b \in S_i} \left\|\mathbf{x}_b - \boldsymbol{\mu}_i\right\|^2 = \sum_{j=1}^{42} w_j^2 \,\mathrm{Var}_{S_i}\!\left(\check{z}_{\cdot,j}\right). \end{aligned}$$

The first term is $\frac{1}{3(\tau_{\mathcal{E}}^R)^2}$ times the sum of squared straight-line (km) distances from each BG's ECEF point to its cluster's mean ECEF point.

Pairwise form. Using $\sum_{\mathbf{x}\in S}\|\mathbf{x}-\boldsymbol{\mu}\|^2 = \frac{1}{2|S|}\sum_{\mathbf{x},\mathbf{y}\in S}\|\mathbf{x}-\mathbf{y}\|^2$, the same objective is

$$\arg\min_{\mathbf{S}} \sum_{i=1}^{k} \frac{1}{2|S_i|} \sum_{b\in S_i}\sum_{b'\in S_i} D^2(b,b'),$$

with the pairwise distance D defined next.

5 The distance function

5.1 General form

$$D(b,b') = \|\mathbf{x}_b - \mathbf{x}_{b'}\|_2, \qquad D^2(b,b') = \sum_{j=1}^{42} w_j^2\big(\check{z}_{b,j} - \check{z}_{b',j}\big)^2 = \big(\tilde{\mathbf{t}}_b - \tilde{\mathbf{t}}_{b'}\big)^\top \Lambda \big(\tilde{\mathbf{t}}_b - \tilde{\mathbf{t}}_{b'}\big),$$

$$\Lambda = \operatorname{diag}\Big(\frac{w_j^2}{(\tau_{[j]}^R)^2}\Big)_{j=1}^{42}.$$

So D is a *diagonally weighted Euclidean distance* on the transformed, imputed features $\tilde{\mathbf{t}}$. Each squared feature difference is divided by its concept's column count and by a regional variance: the feature's own, or the block's pooled variance for ECEF coordinates and share columns. Because whole blocks share one divisor, Λ is not the inverse of the diagonal of the covariance matrix, so D is not a diagonal Mahalanobis distance. There are no off-diagonal terms, so correlations between features are not adjusted for.

5.2 Decomposition by concept

$$D^2(b,b') = \sum_{g\in\mathcal{G}} \Delta_g^2(b,b'), \qquad \Delta_g^2(b,b') = \frac{c_g^2}{n_g} \sum_{j:\, g(j)=g} \Big(\frac{\tilde{t}_{b,j} - \tilde{t}_{b',j}}{\tau_{[j]}^R}\Big)^2.$$

The terms below are written out with all $c_g = 1$, using this notation:

- the superscript R is dropped on τ;
- $\boldsymbol{\pi}_b^K$ is BG b's share vector for block K, and π on its own is a single share;
- τ_K is the block's pooled regional scale;
- $C(v) = \operatorname{clip}(v;\ell_j,u_j)$, using the bounds of the feature it is applied to.

$$\Delta^2_{\text{distance}} = \frac{\|\mathbf{p}_b - \mathbf{p}_{b'}\|_2^2}{3\,\tau_{\mathcal{E}}^2} = \frac{\left(\text{chord}_{bb'}\ \text{km}\right)^2}{3\,\tau_{\mathcal{E}}^2}$$

$$\Delta^2_{\text{race}} = \frac{\left\|\boldsymbol{\pi}^{\text{race}}_b - \boldsymbol{\pi}^{\text{race}}_{b'}\right\|_2^2}{7\,\tau^2_{K_{\text{race}}}} = \frac{1}{7\,\tau^2_{K_{\text{race}}}} \sum_{q\in\{\text{wht,blk,aian,asn,nhpi,oth,hisp}\}} \left(\pi^{\text{race}}_{b,q} - \pi^{\text{race}}_{b',q}\right)^2$$

$$\Delta^2_{\text{density}} = \frac{1}{\tau_\rho^2}\left(\ln\frac{1+\rho_b}{1+\rho_{b'}}\right)^2 \qquad (\rho = \text{people per km}^2)$$

$$\Delta^2_{\text{language}} = \frac{1}{6}\left[\frac{\left\|\boldsymbol{\pi}^{\text{lang}}_b - \boldsymbol{\pi}^{\text{lang}}_{b'}\right\|_2^2}{\tau^2_{K_{\text{lang}}}} + \left(\frac{\pi^{\text{LEP}}_b - \pi^{\text{LEP}}_{b'}}{\tau_{\text{LEP}}}\right)^2\right] \qquad (\boldsymbol{\pi}^{\text{lang}} \in \mathbb{R}^5:\ \text{eng, spa, ie, api, oth})$$

$$\Delta^2_{\text{age}} = \left(\frac{\text{age}_b - \text{age}_{b'}}{\tau_{\text{age}}}\right)^2$$

$$\Delta^2_{\text{income}} = \frac{1}{\tau_I^2}\left(\ln\frac{1+I_b}{1+I_{b'}}\right)^2 \qquad (I = \text{median household income, \$})$$

$$\Delta^2_{\text{dependents}} = \frac{1}{3}\left[\left(\frac{\pi^{\text{kids}}_b - \pi^{\text{kids}}_{b'}}{\tau_{\text{kids}}}\right)^2 + \left(\frac{C(h_b) - C(h_{b'})}{\tau_h}\right)^2 + \frac{1}{\tau^2_{\text{adr}}}\left(\ln\frac{1+C(\text{adr}_b)}{1+C(\text{adr}_{b'})}\right)^2\right]$$

$$\Delta^2_{\text{home type}} = \frac{\left\|\boldsymbol{\pi}^{\text{units}}_b - \boldsymbol{\pi}^{\text{units}}_{b'}\right\|_2^2}{6\,\tau^2_{K_{\text{units}}}} \qquad (\boldsymbol{\pi}^{\text{units}} \in \mathbb{R}^6:\ \text{1det, 1att, 2–4, 5–19, 20+, mobile})$$

$$\Delta^2_{\text{tenure}} = \left(\frac{\pi^{\text{rent}}_b - \pi^{\text{rent}}_{b'}}{\tau_{\text{rent}}}\right)^2$$

$$\Delta^2_{\text{vehicles}} = \frac{1}{2}\left[\left(\frac{\pi^{0\text{veh}}_b - \pi^{0\text{veh}}_{b'}}{\tau_{0\text{veh}}}\right)^2 + \left(\frac{C(v_b) - C(v_{b'})}{\tau_v}\right)^2\right]$$

$$\Delta^2_{\text{commute}} = \frac{1}{\tau_m^2}\left(\ln\frac{1+C(m_b)}{1+C(m_{b'})}\right)^2 \qquad (m = \text{median commute, min})$$

$$\Delta^2_{\text{education}} = \frac{1}{2}\left[\left(\frac{\text{SEDA}_b - \text{SEDA}_{b'}}{\tau_{\text{SEDA}}}\right)^2 + \left(\frac{\text{HUD}_b - \text{HUD}_{b'}}{\tau_{\text{HUD}}}\right)^2\right]$$

$$\Delta^2_{\text{crime}} = \frac{1}{2}\left[\frac{1}{\tau_V^2}\left(\ln\frac{1+C(V_b)}{1+C(V_{b'})}\right)^2 + \frac{1}{\tau_P^2}\left(\ln\frac{1+C(P_b)}{1+C(P_{b'})}\right)^2\right] \qquad (\text{per 100k})$$

$$\Delta^2_{\text{property tax}} = \left(\frac{C(\text{tax}_b) - C(\text{tax}_{b'})}{\tau_{\text{tax}}}\right)^2$$

$$\Delta^2_{\text{occupation}} = \frac{\left\|\boldsymbol{\pi}^{\text{occ}}_b - \boldsymbol{\pi}^{\text{occ}}_{b'}\right\|_2^2}{5\,\tau^2_{K_{\text{occ}}}} \qquad (\boldsymbol{\pi}^{\text{occ}} \in \mathbb{R}^5:\ \text{mgmt, svc, sales, constr, prod})$$

Every quantity here is the imputed value $\tilde{t}$ where the raw value was missing.

5.3 How each kind of term behaves

- **Geography.** $\Delta^2_{\text{distance}}$ is the squared straight-line distance through the Earth between the two population centers, in units of the region's pooled coordinate s.d. Because the three axes share one divisor, the term is rotation-invariant and isotropic: 10 km north counts the same as 10 km east. (Separate per-axis z-scores would multiply N–S and E–W distances by different factors.) On a sphere of radius $\rho_\oplus$, chord $= 2\rho_\oplus \sin(\theta/2)$ for central angle θ, which is within 0.1% of the great-circle distance up to ~ 700 km and increases monotonically with it beyond

that. In the tri-state run, two BGs 50 km apart get $\Delta^2_{\text{distance}} = 50^2/(3 \cdot 94.88^2) \approx 0.093$.

- **Log-transformed features** (density, income, commute, age-dependency, crime). The term depends on the *ratio* $\frac{1+v_b}{1+v_{b'}}$, not the difference, so doubling income adds the same distance at \$30k as at \$150k.
- **Winsorized features.** Differences beyond $[\ell_j, u_j]$ are capped: $|C(v_b) - C(v_{b'})| \le u_j - \ell_j$.
- **Compositional blocks** (race, home type, tract language, occupation) are compared by the plain Euclidean distance between the two share vectors, divided by one pooled scale per block. One percentage point therefore counts the same in every category of a block. Categories that vary more in the region still account for more of the block's variance; for example, the tri-state s.d. of the NH white share is 0.325 but the NHPI share's is 0.005, so rare groups contribute little. Before the fix, each share was z-scored on its own. That divided rare shares by tiny s.d.s, so 1 pp of NHPI counted about 70× as much as 1 pp of NH white, and a few BGs reached $|z| \approx 50$–67. Single shares outside a block (renter, children, zero-vehicle, LEP) are still divided by their own s.d.
- **Within-concept averaging.** Each concept's squared differences are *averaged* over its n_g columns, not summed. That stops a 7-column concept outweighing a 1-column concept.

5.4 Every concept contributes equally on average

Proposition. *Under regional scaling, averaging over all ordered pairs in the region (including $b = b'$),*

$$\frac{1}{n^2} \sum_{b \in \mathcal{B}_R} \sum_{b' \in \mathcal{B}_R} \Delta^2_g(b, b') = 2c_g^2 \quad \forall g, \qquad \frac{1}{n^2} \sum_{b,b'} D^2(b, b') = 2 \sum_g c_g^2 = 30.$$

Proof. For any variable y, $\frac{1}{n^2} \sum_{b,b'} (y_b - y_{b'})^2 = 2\,\mathrm{Var}(y)$.

A feature outside every pooled block has regional variance $\mathrm{Var}(\check{z}_{\cdot,j}) = 1$. For a pooled block P,

$$\sum_{j \in P} \mathrm{Var}(\check{z}_{\cdot,j}) = \frac{\sum_{j \in P} (\tau^R_j)^2}{(\tau^R_P)^2} = |P|.$$

So each concept's columns have total regional variance n_g; the language concept, for example, gets 5 from K_{lang} plus 1 from LEP. Hence $\frac{c_g^2}{n_g} \cdot n_g \cdot 2 = 2c_g^2$. □

So with $c_g = 1$, each of the 15 concepts accounts for 1/15 of the average squared distance. One distance concept's "average" contribution of 2 corresponds to a chord of $\sqrt{6}\,\tau^R_{\mathcal{E}} \approx 232$ km in the tri-state run. Raising c_g to $\sqrt{2}$ doubles concept g's share. This equality does *not* hold under `--scale national`.

6 How the minimization is carried out

`sklearn.cluster.KMeans(n_clusters=k, n_init=25, random_state=seed)` (scikit-learn 1.7; defaults `init="k-means++"`, `algorithm="lloyd"`, `max_iter=300`, `tol=1e-4`) on the $n \times 42$ matrix with rows $\mathbf{x}_b^\top$.

1. **Seeding (greedy k-means++).** Pick $\boldsymbol{\mu}_1^{(0)}$ uniformly from $\{\mathbf{x}_b\}$. Then, for $i = 2, \dots, k$, draw $2 + \lfloor \ln k \rfloor$ candidates with probability $\propto \min_{l<i} \|\mathbf{x}_b - \boldsymbol{\mu}_l^{(0)}\|^2$ and keep the candidate that most reduces the potential.

2. **Lloyd iterations** ($t = 0, 1, \dots$). Assign each BG to its nearest centroid:

$$S_i^{(t)} = \Big\{ b : \ \|\mathbf{x}_b - \boldsymbol{\mu}_i^{(t)}\|^2 \le \|\mathbf{x}_b - \boldsymbol{\mu}_l^{(t)}\|^2 \ \ \forall l \Big\}.$$

   Then update each centroid to its cluster mean:

$$\boldsymbol{\mu}_i^{(t+1)} = \frac{1}{|S_i^{(t)}|} \sum_{b \in S_i^{(t)}} \mathbf{x}_b.$$

   Stop when the assignments stop changing, or when $\sum_i \|\boldsymbol{\mu}_i^{(t+1)} - \boldsymbol{\mu}_i^{(t)}\|^2 \le 10^{-4} \cdot \frac{1}{42} \sum_j \mathrm{Var}(x_{\cdot,j})$, or after 300 iterations. Under regional scaling $\frac{1}{42} \sum_j \mathrm{Var}(x_{\cdot,j}) = \frac{1}{42} \sum_g c_g^2 = 15/42$.

3. **Restarts.** Repeat steps 1–2 25 times and keep the run with the lowest objective (2) (`inertia_`). The result is a local minimum of (1), not a guaranteed global arg min.

4. **Relabeling.** Renumber clusters $1, \dots, k$ so that $|S_1| \ge |S_2| \ge \cdots \ge |S_k|$.

5. **Diagnostic.** The mean silhouette is computed on a uniform sample $\mathcal{U}$ of $\min(20000, n)$ BGs, using the same (unsquared) distance D:

$$s(b) = \frac{\beta(b) - \alpha(b)}{\max\{\alpha(b), \beta(b)\}}, \quad \alpha(b) = \operatorname*{mean}_{b' \in S_{i(b)} \cap \mathcal{U} \setminus \{b\}} D(b, b'), \quad \beta(b) = \min_{l \ne i(b)} \operatorname*{mean}_{b' \in S_l \cap \mathcal{U}} D(b, b').$$

7 Feature table

$w_j = c_{g(j)} / \sqrt{n_{g(j)}}$ with $c_g = 1$. Winsor bounds are in raw units, fitted on $\mathcal{F}$. $s_{[j]}$ is the national scale of $\tilde{t}$ from Step 3.

Table 1: The $d = 42$ features.

j	feature	concept (n_g)	T_j	ℓ_j	u_j	$s_{[j]}$	w_j
1	`ecef_x_km`	distance (3)	none			862.33†	0.5774
2	`ecef_y_km`	distance (3)	none			862.33†	0.5774
3	`ecef_z_km`	distance (3)	none			862.33†	0.5774
4	`race_nh_white_share`	race (7)	none			0.1736†	0.3780
5	`race_nh_black_share`	race (7)	none			0.1736†	0.3780
6	`race_nh_aian_share`	race (7)	none			0.1736†	0.3780
7	`race_nh_asian_share`	race (7)	none			0.1736†	0.3780
8	`race_nh_nhpi_share`	race (7)	none			0.1736†	0.3780
9	`race_nh_other_multi_share`	race (7)	none			0.1736†	0.3780
10	`race_hispanic_share`	race (7)	none			0.1736†	0.3780
11	`pop_density_per_sqkm`	density (1)	log1p			2.0590	1.0000
12	`lang_tract_english_only_share`	language (6)	none			0.1305†	0.4082
13	`lang_tract_spanish_share`	language (6)	none			0.1305†	0.4082
14	`lang_tract_other_ie_share`	language (6)	none			0.1305†	0.4082
15	`lang_tract_asian_pi_share`	language (6)	none			0.1305†	0.4082

j	feature	concept (n_g)	T_j	ℓ_j	u_j	$s_{[j]}$	w_j
16	`lang_tract_other_share`	language (6)	none			0.1305†	0.4082
17	`lep_hh_share`	language (6)	none			0.0863	0.4082
18	`median_age`	age (1)	none			10.270	1.0000
19	`median_hh_income`	income (1)	log1p			0.5245	1.0000
20	`hh_with_children_share`	dependents (3)	none			0.1543	0.5774
21	`avg_hh_size`	dependents (3)	winsor	1.29	4.77	0.6117	0.5774
22	`age_dependency_ratio`	dependents (3)	winsor_log1p	0.0349	3.0321	0.1916	0.5774
23	`units_1_detached_share`	home type (6)	none			0.1816†	0.4082
24	`units_1_attached_share`	home type (6)	none			0.1816†	0.4082
25	`units_2_4_share`	home type (6)	none			0.1816†	0.4082
26	`units_5_19_share`	home type (6)	none			0.1816†	0.4082
27	`units_20plus_share`	home type (6)	none			0.1816†	0.4082
28	`units_mobile_other_share`	home type (6)	none			0.1816†	0.4082
29	`renter_share`	tenure (1)	none			0.2733	1.0000
30	`zero_vehicle_hh_share`	vehicles (2)	none			0.1359	0.7071
31	`mean_vehicles_per_hh`	vehicles (2)	winsor	0.2137	3.0653	0.5207	0.7071
32	`median_commute_min`	commute (1)	winsor_log1p	7.8486	62.320	0.3429	1.0000
33	`school_quality_index`	education (2)	none			0.3359	0.7071
34	`hud_school_proficiency_index`	education (2)	none			28.692	0.7071
35	`violent_crime_rate`	crime (2)	winsor_log1p	16.103	1649.1	0.9922	0.7071
36	`property_crime_rate`	crime (2)	winsor_log1p	54.774	5155.7	0.8896	0.7071
37	`eff_property_tax_rate_tract`	property tax (1)	winsor	0.001875	0.029171	0.005522	1.0000
38	`occ_mgmt_prof_share`	occupation (5)	none			0.1283†	0.4472
39	`occ_service_share`	occupation (5)	none			0.1283†	0.4472
40	`occ_sales_office_share`	occupation (5)	none			0.1283†	0.4472
41	`occ_construction_maint_share`	occupation (5)	none			0.1283†	0.4472
42	`occ_production_transp_share`	occupation (5)	none			0.1283†	0.4472

†Pooled block scale s_P, shared by every column of the block: ECEF (km), race, home-type units, tract language and occupation. Under regional scaling, the national $s_{[j]}$ are replaced by the regional $\tau^R_{[j]}$. For CT+NJ+NY the pooled values are $\tau^R_{\mathcal{E}} = 94.88$ km, $\tau^R_{K_{\text{race}}} = 0.1764$, $\tau^R_{K_{\text{units}}} = 0.2219$, $\tau^R_{K_{\text{lang}}} = 0.1416$ and $\tau^R_{K_{\text{occ}}} = 0.1280$.

8 Data sources

All inputs are public and are downloaded by the feature modules (`features/*.py`). Feature numbers refer to Table 1. SEDA data are for non-commercial use; cite SEDA and do not redistribute its raw files.

Table 2: Data sources (each name links to the source).

Source	Vintage	Native geography	Used for
American Community Survey 5-year estimates (U.S. Census Bureau, Census Data API)	2019–2023	block group; tract for C16001 and the property-tax tables	features 4–32 and 37–42 (tables B03002, B01003, C16001, C16002, B01002, B19013, B11005, B25010, B01001, B25024, B25003, B25044, B08303, B25090, B25082, B25103, B25077, C24010); population denominators for density and crime rates
TIGER/Line block groups (U.S. Census Bureau)	2023	block group	land area for feature 11; fallback location for 1–3 (about 0.6% of block groups); map polygons
2020 Census centers of population (U.S. Census Bureau)	2020	block group	features 1–3 (population-weighted center); locating each block group for 33–36
Cartographic boundary files, 1:500,000 (county, place, county subdivision; U.S. Census Bureau)	2023	county, place, county subdivision	police jurisdictions for 35–36; shoreline clipping of map polygons
Stanford Education Data Archive (SEDA) 6.0, geographic-district file and district shapefile (Reardon et al.)	school years 2008–09 to 2018–19	school district	feature 33
SEDA 2025.2, administrative-district file	same period	school district	feature 33 for districts missing from SEDA 6.0 (DC, HI and a few others)
HUD AFFH School Proficiency Index (version AFFHT0006)	2015–16 school year	2010 block group (attendance zones)	feature 34
FBI Crime Data Explorer, Offenses Known to Law Enforcement, Tables 8 and 10	data year 2025	police agency (city or county)	features 35–36
County Health Rankings analytic data (violent crime, UCR 2014 and 2016)	2022 release	county	features 35–36 where no usable FBI data exist